

# Verification

## o2\_H\_Simulator\_FVM\_1D\_Incompressible\_Heterogeneous

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### 1 Objective

The purpose of this verification exercise is to demonstrate that the numerical simulator correctly reproduces the analytical solution for one-dimensional incompressible single-phase flow in a heterogeneous porous medium.

### 2 Governing Equation

For steady-state incompressible flow with no gravity and no source terms, conservation of mass combined with Darcy's law gives

$$\frac{\partial}{\partial x} \left( -\frac{k(x)}{\mu} \frac{\partial P}{\partial x} \right) = 0, \quad (1)$$

where

- $P$  is pressure [Pa]
- $k$  is permeability [ $\text{m}^2$ ]
- $\mu$  is fluid viscosity [ $\text{Pa}\cdot\text{s}$ ]

The permeability field consists of two homogeneous regions,

$$k(x) = \begin{cases} k_1, & 0 \leq x < L_1, \\ k_2, & L_1 \leq x \leq L_x, \end{cases} \quad (2)$$

where  $L_1$  is the location of the interface between the two permeability zones and  $L_x$  is the total domain length. The boundary conditions are

$$P(0) = P_L, \quad P(L_x) = P_R. \quad (3)$$

### 3 Analytical Solution

Because the flow is steady state and incompressible, the Darcy velocity is constant throughout the domain,

$$u = -\frac{k_1}{\mu} \frac{dP}{dx} = -\frac{k_2}{\mu} \frac{dP}{dx}. \quad (4)$$

The pressure solution is therefore piecewise linear,

$$P(x) = P_L - \frac{\mu u}{k_1} x, \quad 0 \leq x \leq L_1, \quad (5)$$

and

$$P(x) = P(L_1) - \frac{\mu u}{k_2} (x - L_1), \quad L_1 \leq x \leq L_x. \quad (6)$$

## 4 Effective Permeability

For steady-state incompressible flow, Darcy's law in one dimension is

$$Q = -\frac{kA}{\mu} \frac{dP}{dx}, \quad (7)$$

where  $Q$  is the volumetric flow rate. Since the flow is steady state and there are no sources or sinks,  $Q$  is constant throughout the domain. Consider a domain composed of two regions in series,

$$L_x = L_1 + L_2, \quad (8)$$

with permeabilities  $k_1$  and  $k_2$ , respectively. The pressure drop across each region is

$$\Delta P_1 = \frac{\mu Q L_1}{A k_1}, \quad (9)$$

and

$$\Delta P_2 = \frac{\mu Q L_2}{A k_2}. \quad (10)$$

The total pressure drop across the domain is therefore

$$P_L - P_R = \Delta P_1 + \Delta P_2 = \frac{\mu Q}{A} \left( \frac{L_1}{k_1} + \frac{L_2}{k_2} \right). \quad (11)$$

An equivalent homogeneous permeability,  $k_{\text{eff}}$ , can be defined such that

$$Q = \frac{k_{\text{eff}} A}{\mu} \frac{P_L - P_R}{L_x}. \quad (12)$$

Substituting the expression for the total pressure drop yields

$$\frac{L_x}{k_{\text{eff}}} = \frac{L_1}{k_1} + \frac{L_2}{k_2}, \quad (13)$$

and therefore

$$k_{\text{eff}} = \frac{L_x}{\frac{L_1}{k_1} + \frac{L_2}{k_2}}. \quad (14)$$

For the special case in which the two regions have equal length,

$$L_1 = L_2 = \frac{L_x}{2}, \quad (15)$$

the effective permeability reduces to

$$k_{\text{eff}} = \frac{2k_1 k_2}{k_1 + k_2}, \quad (16)$$

which is the harmonic average of the two permeabilities. The numerical solution obtained using the heterogeneous permeability field can therefore be compared against a homogeneous simulation using  $k_{\text{eff}}$ . Agreement between the two flow rates validates the implementation of heterogeneous permeability and transmissibility calculations.

## 5 Simulation Parameters

The simulation parameters are given in Table 1.

| Parameter       | Symbol | Value               |
|-----------------|--------|---------------------|
| Number of cells | $N_x$  | 100                 |
| Domain length   | $L_x$  | 1 m                 |
| Zone 1 length   | $L_1$  | 0.5 m               |
| Permeability 1  | $k_1$  | 1 m <sup>2</sup>    |
| Permeability 2  | $k_2$  | 5 m <sup>2</sup>    |
| Viscosity       | $\mu$  | 1 Pa·s              |
| Left pressure   | $P_L$  | $10 \times 10^6$ Pa |
| Right pressure  | $P_R$  | 0 Pa                |

Table 1: Simulation parameters.

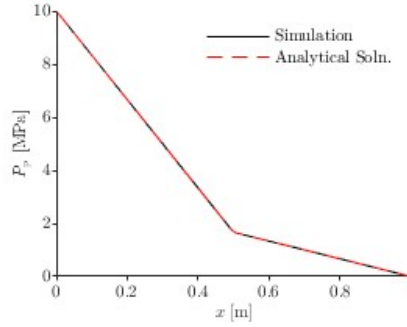


Figure 1: Comparison between analytical and numerical pressure solutions for the MATLAB simulator.

## 6 Results

### 6.1 Pressure Distribution

The pressure solutions for both the MATLAB simulator, Figure 1, and the Python simulator, Figure ??, match well.

### 6.2 Darcy Flux

The Darcy flux solutions for both the MATLAB simulator, Figure 2, and the Python simulator, Figure ??, match well.

## 7 Error Analysis

The maximum pressure relative error is defined as

$$E_P = \max_i \left| \frac{P_{\text{num},i} - P_{\text{ana},i}}{q_{\text{ana},i}} \right|. \quad (17)$$

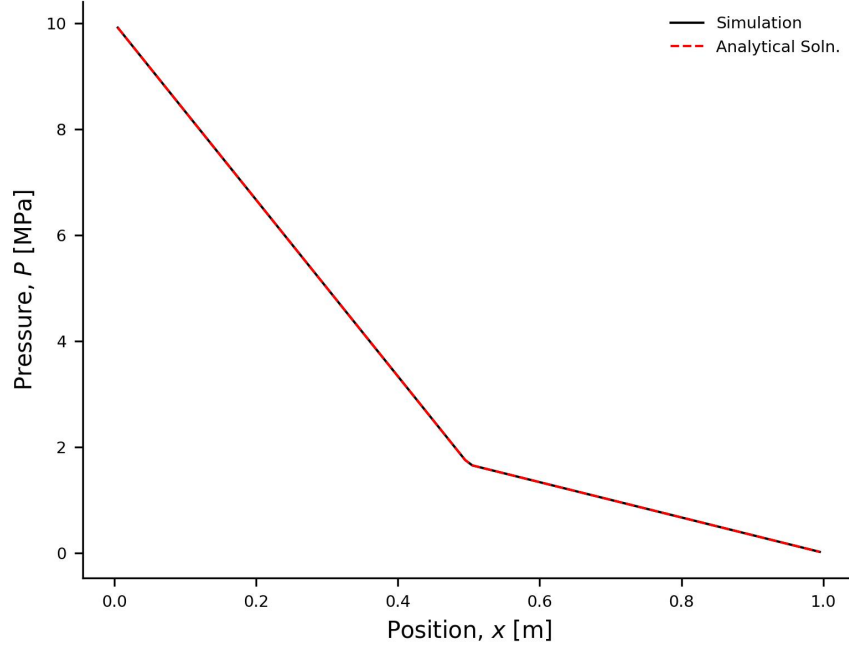


Figure 2: Comparison between analytical and numerical pressure solutions for the Python simulator.

The maximum flux relative error is defined as

$$E_q = \max_i \left| \frac{q_{\text{num},i} - q_{\text{ana},i}}{q_{\text{ana},i}} \right|. \quad (18)$$

For the present simulation in MATLAB

$$E_P = 4.60 \times 10^{-14}, \quad E_q = 5.21 \times 10^{-14}. \quad (19)$$

In Python, the result is

$$E_P = 3.55 \times 10^{-14}, \quad E_q = 3.96 \times 10^{-14}. \quad (20)$$

## 8 Discussion

The numerical pressure distribution reproduces the analytical linear pressure solution and the numerical Darcy flux reproduces the analytical constant flux solution. The small residual errors arise from floating-point arithmetic and are negligible.

## 9 Conclusion

The simulator successfully reproduces the analytical solution for one-dimensional incompressible flow in a heterogeneous porous medium. This provides verification that the finite volume implementation is correctly solving the governing equations for this benchmark problem.

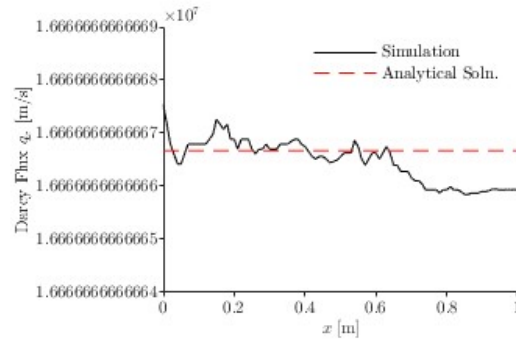


Figure 3: Comparison between analytical and numerical Darcy flux solutions for the MATLAB simulator.

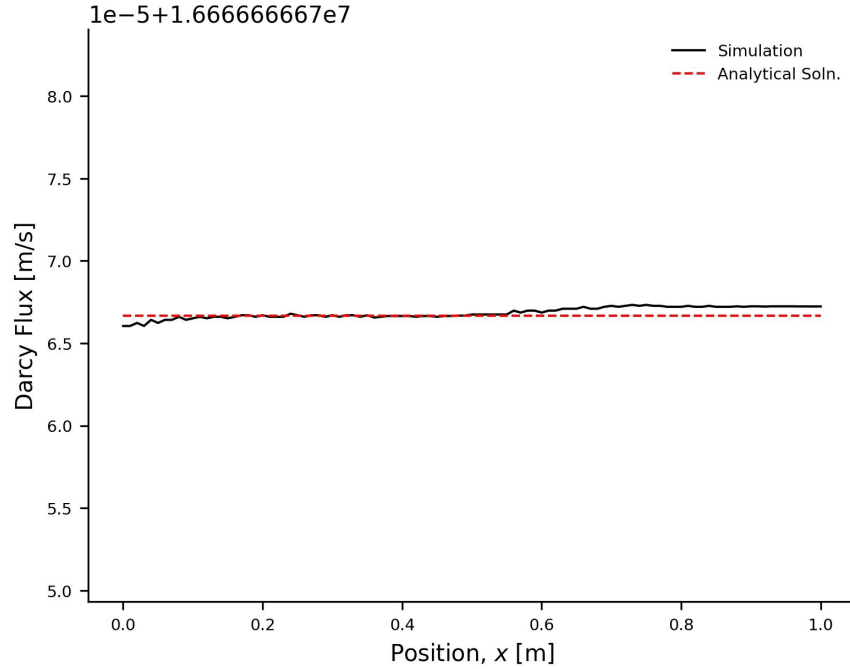


Figure 4: Comparison between analytical and numerical Darcy flux solutions for the Python simulator.